

# Software Reference Manual

Other specifications without this document conform to the IMX678 Datasheet.

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## Description

In this specification, it is list about “Operating mode” and “Description of Various Function”.  
Please refer to the “Datasheet” and “IMX678\_Stadard\_Register\_Setting (Excel file)” for matters not described in this document.

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## Features

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- ◆ Standby Mode
- ◆ Slave Mode and Master Mode
- ◆ Gain Adjustment Function
- ◆ Black Level Adjustment Function
- ◆ Normal Operation and Inverted Operation
- ◆ Shutter and Integration Time Setting
- ◆ Normal Exposure Operation
- ◆ Long Exposure Operation
- ◆ Example of Integration Time Settings
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- ◆ CSI-2 Output
- ◆ MIPI Transmitter
- ◆ INCK Setting
- ◆ Global Timing setting
- ◆ Register Hold Setting
- ◆ Mode Transitions
- ◆ XSIZE\_OVERLAP Setting

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## Operating mode

The table below shows the operating modes available with this sensor.

| Mode      | Lane  | Data rate<br>[Mbps/<br>Lane] | AD<br>conversion<br>[bit] | Output<br>bit width<br>[bit] | Frame<br>rate<br>[frame/s] | Recording Pixels |              | INCK<br>[MHz]                                                | 1H period<br>[Clock] <sup>(*)</sup> | 1V period<br>[XHS] |
|-----------|-------|------------------------------|---------------------------|------------------------------|----------------------------|------------------|--------------|--------------------------------------------------------------|-------------------------------------|--------------------|
|           |       |                              |                           |                              |                            | H<br>[pixels]    | V<br>[lines] |                                                              |                                     |                    |
| All pixel | 2     | 1440                         | 10                        | 10                           | 25.0                       | 3840             | 2160         | 13.5,<br>18,<br>24,<br>27,<br>36,<br>37.125,<br>72,<br>74.25 | 1320                                | 2250               |
|           |       | 1782                         | 10                        | 10                           | 30.0                       |                  |              |                                                              | 1100                                |                    |
|           |       | 1782                         | 12                        | 12                           | 25.0                       |                  |              |                                                              | 1320                                |                    |
|           |       | 2079                         | 12                        | 12                           | 30.0                       |                  |              |                                                              | 1100                                |                    |
|           | 4     | 720                          | 10                        | 10                           | 25.0                       |                  |              |                                                              | 1320                                |                    |
|           |       | 891                          | 10                        | 10                           | 30.0                       |                  |              |                                                              | 1100                                |                    |
|           |       | 891                          | 12                        | 12                           | 25.0                       |                  |              |                                                              | 1320                                |                    |
|           |       | 1188                         | 12                        | 12                           | 30.0                       |                  |              |                                                              | 1100                                |                    |
|           |       | 1440                         | 10                        | 10                           | 50.0                       |                  |              |                                                              | 660                                 |                    |
|           |       | 1440                         | 10                        | 10                           | 60.0                       |                  |              |                                                              | 550                                 |                    |
|           |       | 1440                         | 12                        | 12                           | 50.0                       |                  |              |                                                              | 660                                 |                    |
|           |       | 1782                         | 12                        | 12                           | 60.0                       |                  |              |                                                              | 550                                 |                    |
|           |       | 2079                         | 10                        | 10                           | 72.0                       |                  |              |                                                              | 458                                 |                    |
|           | 8     | 525                          | 10                        | 10                           | 25.0                       |                  |              | 36,<br>72                                                    | 1320                                |                    |
|           |       | 525                          | 10                        | 10                           | 30.0                       |                  |              |                                                              | 1100                                |                    |
|           |       | 525                          | 12                        | 12                           | 25.0                       |                  |              |                                                              | 1320                                |                    |
|           |       | 525                          | 12                        | 12                           | 30.0                       |                  |              |                                                              | 1100                                |                    |
|           |       | 594                          | 10                        | 10                           | 25.0                       |                  |              | 13.5,<br>18,<br>24,<br>27,<br>36,<br>37.125,<br>72,<br>74.25 | 1320                                |                    |
|           |       | 594                          | 10                        | 10                           | 30.0                       |                  |              |                                                              | 1100                                |                    |
|           |       | 594                          | 12                        | 12                           | 25.0                       |                  |              |                                                              | 1320                                |                    |
|           |       | 594                          | 12                        | 12                           | 30.0                       |                  |              |                                                              | 1100                                |                    |
|           |       | 720                          | 10                        | 10                           | 50.0                       |                  |              |                                                              | 660                                 |                    |
|           |       | 891                          | 10                        | 10                           | 60.0                       |                  |              |                                                              | 550                                 |                    |
|           |       | 891                          | 12                        | 12                           | 50.0                       |                  |              |                                                              | 660                                 |                    |
|           |       | 1188                         | 12                        | 12                           | 60.0                       |                  |              |                                                              | 550                                 |                    |
|           |       | 1188                         | 10                        | 10                           | 72.0                       |                  |              |                                                              | 458                                 |                    |
|           | 4×2ch | 525                          | 10                        | 10                           | 25.0                       |                  |              | 36,<br>72                                                    | 1320                                |                    |
|           |       | 525                          | 10                        | 10                           | 30.0                       |                  |              |                                                              | 1100                                |                    |
|           |       | 525                          | 12                        | 12                           | 25.0                       |                  |              |                                                              | 1320                                |                    |
|           |       | 525                          | 12                        | 12                           | 30.0                       |                  |              |                                                              | 1100                                |                    |
|           |       | 594                          | 10                        | 10                           | 25.0                       |                  |              | 13.5,<br>18,<br>24,<br>27,<br>36,<br>37.125,<br>72,<br>74.25 | 1320                                |                    |
|           |       | 594                          | 10                        | 10                           | 30.0                       |                  |              |                                                              | 1100                                |                    |
|           |       | 594                          | 12                        | 12                           | 25.0                       |                  |              |                                                              | 1320                                |                    |
|           |       | 594                          | 12                        | 12                           | 30.0                       |                  |              |                                                              | 1100                                |                    |
|           |       | 720                          | 10                        | 10                           | 50.0                       |                  |              |                                                              | 660                                 |                    |
|           |       | 891                          | 10                        | 10                           | 60.0                       |                  |              |                                                              | 550                                 |                    |
|           |       | 891                          | 12                        | 12                           | 50.0                       |                  |              |                                                              | 660                                 |                    |
|           |       | 1188                         | 12                        | 12                           | 60.0                       |                  |              |                                                              | 550                                 |                    |
|           |       | 1188                         | 10                        | 10                           | 72.0                       |                  |              |                                                              | 458                                 |                    |

| Mode                                           | Lane  | Data rate<br>[Mbps/<br>Lane] | AD<br>conversion<br>[bit] | Output<br>bit width<br>[bit] | Frame<br>rate<br>[frame/s] | Recording Pixels |              | INCK<br>[MHz]                                                | 1H period<br>[Clock] <sup>(*)</sup> | 1V period<br>[XHS] |
|------------------------------------------------|-------|------------------------------|---------------------------|------------------------------|----------------------------|------------------|--------------|--------------------------------------------------------------|-------------------------------------|--------------------|
|                                                |       |                              |                           |                              |                            | H<br>[pixels]    | V<br>[lines] |                                                              |                                     |                    |
| Horizontal/<br>Vertical<br>2/2-line<br>binning | 2     | 1440                         | 10                        | 12                           | 25.0                       | 1920             | 1080         | 13.5,<br>18,<br>24,                                          | 1320                                | 2250               |
|                                                |       | 1440                         | 10                        | 12                           | 30.0                       |                  |              |                                                              | 1100                                |                    |
|                                                | 4     | 720                          | 10                        | 12                           | 25.0                       |                  |              | 27,<br>36,<br>37.125,<br>72,<br>74.25                        | 1320                                |                    |
|                                                |       | 720                          | 10                        | 12                           | 30.0                       |                  |              |                                                              | 1100                                |                    |
|                                                |       | 1440                         | 10                        | 12                           | 50.0                       |                  |              |                                                              | 660                                 |                    |
|                                                |       | 1440                         | 10                        | 12                           | 60.0                       |                  |              |                                                              | 550                                 |                    |
|                                                |       | 2376                         | 10                        | 12                           | 72.0                       |                  |              |                                                              | 458                                 |                    |
|                                                | 8     | 525                          | 10                        | 12                           | 25.0                       |                  |              | 36,<br>72                                                    | 1320                                |                    |
|                                                |       | 525                          | 10                        | 12                           | 30.0                       |                  |              |                                                              | 1100                                |                    |
|                                                |       | 594                          | 10                        | 12                           | 25.0                       |                  |              | 13.5,<br>18,<br>24,<br>27,<br>36,<br>37.125,<br>72,<br>74.25 | 1320                                |                    |
|                                                |       | 594                          | 10                        | 12                           | 30.0                       |                  |              |                                                              | 1100                                |                    |
|                                                |       | 720                          | 10                        | 12                           | 50.0                       |                  |              |                                                              | 660                                 |                    |
|                                                |       | 891                          | 10                        | 12                           | 60.0                       |                  |              |                                                              | 550                                 |                    |
|                                                |       | 1440                         | 10                        | 12                           | 72.0                       |                  |              |                                                              | 458                                 |                    |
|                                                | 4×2ch | 525                          | 10                        | 12                           | 25.0                       |                  |              | 36,<br>72                                                    | 1320                                |                    |
|                                                |       | 525                          | 10                        | 12                           | 30.0                       |                  |              |                                                              | 1100                                |                    |
|                                                |       | 594                          | 10                        | 12                           | 25.0                       |                  |              | 13.5,<br>18,<br>24,<br>27,<br>36,<br>37.125,<br>72,<br>74.25 | 1320                                |                    |
|                                                |       | 594                          | 10                        | 12                           | 30.0                       |                  |              |                                                              | 1100                                |                    |
|                                                |       | 720                          | 10                        | 12                           | 50.0                       |                  |              |                                                              | 660                                 |                    |
|                                                |       | 891                          | 10                        | 12                           | 60.0                       |                  |              |                                                              | 550                                 |                    |
|                                                |       | 1440                         | 10                        | 12                           | 72.0                       |                  |              |                                                              | 458                                 |                    |
|                                                |       | 1440                         | 10                        | 12                           | 72.0                       |                  |              |                                                              | 458                                 |                    |

(\*) Clock frequency = 74.25 [MHz]

### Frame rate on All-pixel and Horizontal / Vertical 2/2-line binning

The table below shows the register setting example of typical frame rate.

The frame rate is obtained by the following formula when using other frame rates.

Frame rate [frame / s] = 1 / ( V<sub>TTL</sub> × (1H period) )

V<sub>TTL</sub> : 1 frame line length or VMAX  
: "1V period" or more in "Operating mode"  
1H period (unit [s]) : "1H period" or more in "Operating mode"

## Description of Various Function

### Window Cropping Mode

Sensor signals are cut out and read out in arbitrary positions.

This function support All-pixel mode, Horizontal/Vertical 2/2-line binning mode, Digital overlap HDR, Clear HDR and Vertical / Horizontal direction-normal / inverted readout mode of each modes.

Cropping position is set, regarding effective pixel start position as origin (0, 0) in normal mode direction. That is a start point which is an offset from the origin and cropping width.

Cropping is available from all-pixel scan mode and horizontal period is fixed to the value for this mode. Pixels cropped by horizontal cropping setting are output with left shifted and that extends the horizontal blanking period. Window position and size is used fixed value. (An ignore frame is output when it is changed.)

Window cropping image is shown in the figure below.

The same physical pixel area as all-pixel mode is cropped when start position and width are same setting in Horizontal/Vertical 2/2-line binning mode, Digital overlap HDR, Clear HDR and Vertical / Horizontal direction-normal / inverted readout mode.

At inverted mode, it is the same as the "Recording pixel with Effective margin for color processing (green rectangle in the figure)" area in normal mode.

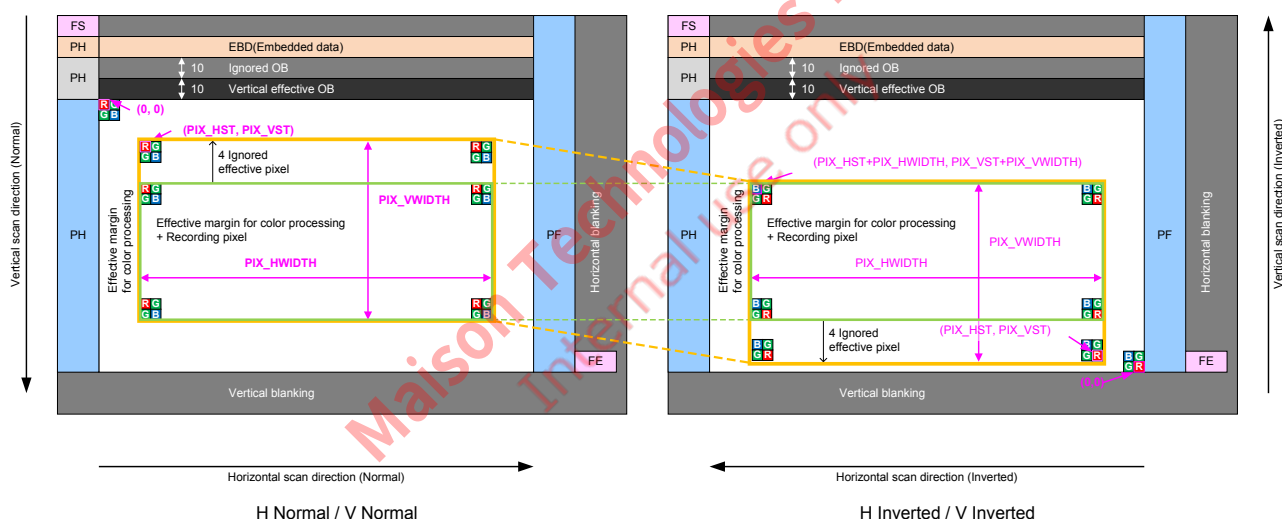


Image Drawing of Window Cropping Mode in Horizontal/Vertical, normal/inverted direction

List of Setting Register

| Register   | Register details |       | Initial value | Setting value                                         | Remarks               |
|------------|------------------|-------|---------------|-------------------------------------------------------|-----------------------|
|            | Address          | bit   |               |                                                       |                       |
| WINMODE    | 3018h            | [3:0] | 00h           | 04h: Window Cropping mode                             |                       |
| PIX_HST    | 303Ch            | [7:0] | 0000h         | Effective pixel Start position (Horizontal direction) | Set a multiple of 2.  |
|            | 303Dh            | [4:0] |               |                                                       |                       |
| PIX_HWIDTH | 303Eh            | [7:0] | 0F10h         | Effective pixel Cropping width (Horizontal direction) | Set a multiple of 16. |
|            | 303Fh            | [4:0] |               |                                                       |                       |
| PIX_VST    | 3044h            | [7:0] | 0000h         | Effective pixel Start position (Vertical direction)   | Set a multiple of 4.  |
|            | 3045h            | [3:0] |               |                                                       |                       |
| PIX_VWIDTH | 3046h            | [7:0] | 0884h         | Effective pixel Cropping width (Vertical direction)   | Set a multiple of 4.  |
|            | 3047h            | [3:0] |               |                                                       |                       |

**Restrictions on Window cropping mode**

The register settings should satisfy following conditions:

Set WINMODE: 4h.

◆ PIX\_VST, PIX\_VWIDTH

Set PIX\_VST, PIX\_VWIDTH to a multiple of 4.

$$\begin{aligned} \text{PIX\_VST} &= n_1 \times 4 & (n_1 > 0) \\ \text{PIX\_VWIDTH} &= n_2 \times 4 & (n_2 \geq 239) \end{aligned}$$

Cropped area is needed to set pre 4 pixel, rear 0 pixel for signal processing.

◆ PIX\_HST, PIX\_HWIDTH

Set PIX\_HST to a multiple of 2.

Set PIX\_HWIDTH to a multiple of 16.

$$\begin{aligned} \text{PIX\_HST} &= n_3 \times 2 & (n_3 > 0) \\ \text{PIX\_HWIDTH} &= n_4 \times 16 & (n_4 > 64, n_4 \neq 128) \end{aligned}$$

Where  $n_1$  to 4 are integer equal or more than 0.

$V_{\text{TTL}}$  (1frame line length or VMAX)  $\geq$  PIX\_VWIDTH + 70

Set  $V_{\text{TTL}}$  to 1024 or more.

$$V_{\text{TTL}} \geq 1024$$

◆ Frame rate on Window cropping mode

Frame rate [frame/s] =  $1 / (V_{\text{TTL}} \times (1\text{H period}))$

1H period (unit:  $\mu\text{s}$ ) : Fix 1H time in a mode before cropping and refer to the value of "1H period" in the table of "Operating mode".

Where  $V_{\text{TTL}}$  is 1frame line length or VMAX.

## Standby Mode

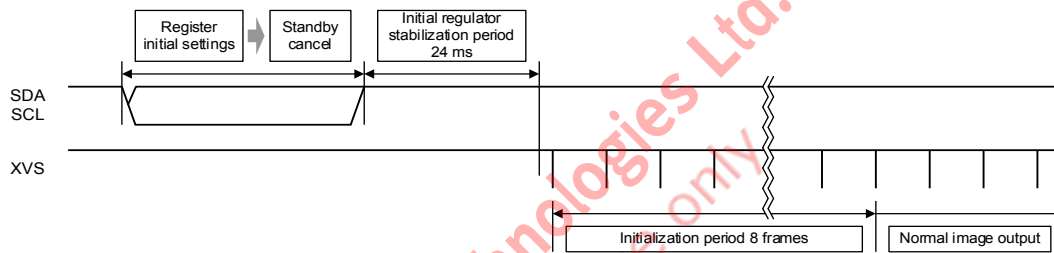
This sensor stops its operation and goes into standby mode which reduces the power consumption by writing “1” to the standby control register STANDBY. Standby mode is also established after power-on or other system reset operation.

### List of Standby Mode Setting

| Register | Register details |     | Initial value | Setting value                | Remarks                                             |
|----------|------------------|-----|---------------|------------------------------|-----------------------------------------------------|
|          | Address          | bit |               |                              |                                                     |
| STANDBY  | 3000h            | [0] | 1h            | 1h: Standby<br>0h: Operating | Register communication is executed in standby mode. |

The serial communication registers hold the previous values. However, the address registers transmitted in standby mode are overwritten. The serial communication block operates even in standby mode, so standby mode can be canceled by setting the STANDBY register to “0”. Some time is required for sensor internal circuit stabilization after standby mode is canceled. After standby mode is canceled, a normal image is output from the 8 frames after internal regulator stabilization 24 ms or more.

For details of the sequence of setting and cancel standby mode, see the sensor setting flow after power on.



Sequence from Standby Cancel to Stable Image Output

## Slave Mode and Master Mode

The sensor can be switched between slave mode and master mode. The switching is made by the XMASTER pin. Establish the XMASTER pin status before canceling the system reset. (Do not switch this register status during operation.)

Input a vertical sync signal to XVS and input a horizontal sync signal to XHS when a sensor is in slave mode. For sync signal interval, input data lines to output for vertical sync signal and 1H period designated in each operating mode for horizontal sync signal. See the section of "Operating mode" for the number of output data line and 1H period.

Set the XMSTA register 0h to start the operation after setting to master mode. In addition, set the count number of sync signal in vertical direction by the VMAX [19:0] register and the clock number in horizontal direction by the HMAX [15:0] register. See the description of Operation mode for details of the section of "Operating mode".

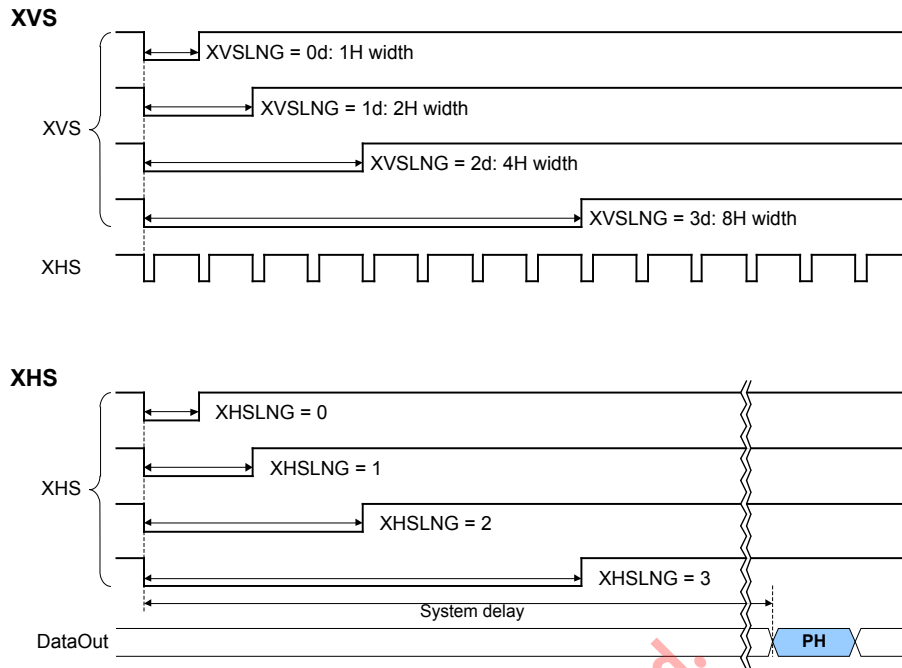
### List of Slave and Master Mode Setting

| Pin name | Pin processing | Setting value | Remarks                |
|----------|----------------|---------------|------------------------|
| XMASTER  | Fixed to Low   | Master mode   | High: OV <sub>DD</sub> |
|          | Fixed to High  | Slave mode    | Low: GND               |

### List of Register in Master Mode

| Register        | Register details |       | Initial value | Setting value                                                              | Remarks                                                       |
|-----------------|------------------|-------|---------------|----------------------------------------------------------------------------|---------------------------------------------------------------|
|                 | Address          | bit   |               |                                                                            |                                                               |
| XMSTA           | 3002h            | [0]   | 1h            | 0h: Master mode operation start<br>1h: Master mode operation stop          | The master operation starts by setting 0.                     |
| VMAX [19:0]     | 3028h            | [7:0] | 008CAh        | See the item of each drive mode.                                           | Line number per frame designated<br>* Set value multiple of 2 |
|                 | 3029h            | [7:0] |               |                                                                            |                                                               |
|                 | 302Ah            | [3:0] |               |                                                                            |                                                               |
| HMAX [15:0]     | 302Ch            | [7:0] | 044Ch         | See the item of each drive mode.                                           | Clock number per line designated                              |
|                 | 302Dh            | [7:0] |               |                                                                            |                                                               |
| XVSOUTSEL [1:0] | 30A4h            | [1:0] | 2h            | 0h: Fixed to Low<br>2h: VSYNC output                                       | -                                                             |
| XHSOUTSEL [1:0] |                  | [3:2] | 2h            | 0h: Fixed to Low<br>2h: HSYNC output                                       | -                                                             |
| XVS_DRV [1:0]   | 30A6h            | [1:0] | 3h            | 0h: XVS output (Master mode)<br>3h: Hi-z (Slave mode)                      | -                                                             |
| XHS_DRV [1:0]   |                  | [3:2] | 3h            | 0h: XHS output (Master mode)<br>3h: Hi-z (Slave mode)                      | -                                                             |
| XVSLNG [1:0]    | 30CCh            | [5:4] | 0h            | 0h: 1H, 1h: 2H, 2h: 4H, 3h: 8H                                             | XVS low level pulse width designated                          |
| XHSLNG [1:0]    | 30CDh            | [5:4] | 0h            | 0h: 16clock, 1h: 32clock<br>2h: 64clock, 3h: 128clock<br>See the next page | XHS low level pulse width designated                          |





XVS/XHS output waveform in sensor master mode

The XVS and XHS are output in timing that set 0 to the register XMSTA. If set 0 to XMSTA during standby, the XVS and XHS are output just after standby is released. The XVS and XHS are output asynchronous with other input or output signals. In addition, the output signals are output with an undefined latency time (system delay) relative to the XHS. Therefore, refer to the sync codes output from the sensor and perform synchronization.

## Gain Adjustment Function

The Programmable Gain Control (PGC) of this device consists of the analog block and digital block. The total of analog gain and digital gain can be set up to 72dB by the GAIN [10:0] register setting. The same setting is applied in all colors.

The value which is 10/3 times the gain is set to register. (0.3 dB step)

Example)

When set to 6 dB:  $6 \times 10/3 = 20d$ ; GAIN = 14h

When set to 12.6 dB:  $12.6 \times 10/3 = 42d$ ; GAIN = 2Ah

### List of PGC Register

| Register | Register details |       | Initial value | Setting value              | Remarks                                                 |
|----------|------------------|-------|---------------|----------------------------|---------------------------------------------------------|
|          | Address          | bit   |               | Setting range              |                                                         |
| GAIN_0   | 3070h            | [7:0] | 000h          | 00h to F0h<br>(0d to 240d) | Setting value: Gain [dB] $\times$ 10/3<br>(0.3 dB step) |
|          | 3071h            | [2:0] |               |                            |                                                         |

When using conversion gain high (FDG\_SEL0 = 1), there is following restriction on setting range of Gain.

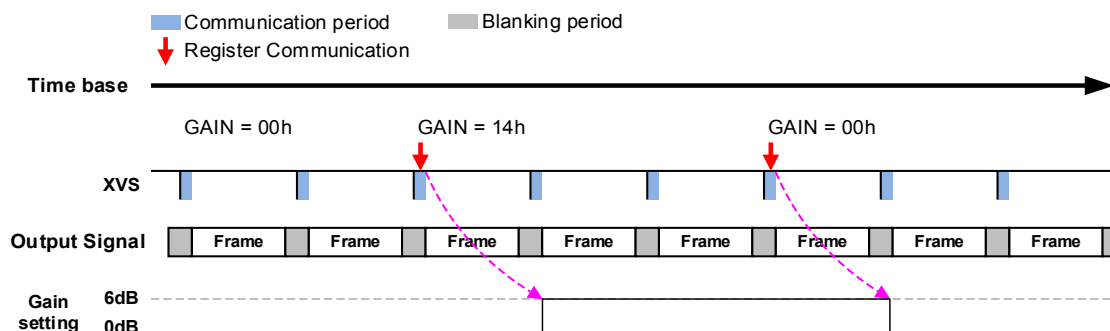
### List of Conversion Gain Register

| Register       | Register details |       | Initial value | Setting value | Remarks                                      |
|----------------|------------------|-------|---------------|---------------|----------------------------------------------|
|                | Address          | bit   |               | Setting range |                                              |
| FDG_SEL0 [1:0] | 3030h            | [1:0] | 0h            | 0h: LCG       | Gain setting range: 00h to F0h (0d to 240d)  |
|                |                  |       |               | 1h: HCG       | Gain setting range: 22h to F0h (34d to 240d) |

Followings are other notations when changing conversion gain by the register.

1. Saturation signal gets smaller according to Vsat in "Image Sensor Characteristics" of Datasheet.

The gain setting is reflected at the next frame that the communication is performed as shown below.



Gain Reflection Timing

## Black Level Adjustment Function

The black level offset (offset variable range: 000h to 3FFh) can be added relative to the data in which the digital gain modulation was performed by the BLKLEVEL [9:0] register.

Note that the offset unit changes according to the output bit setting.

When the output data length is 10-bit output, increasing the register setting value by 1h increases the black level by 1 LSB. When the output data length is 12-bit output, increasing the register setting value by 1h increases the black level by 4 LSB.

Use with values shown below is recommended.

10-bit output: 032h (50d in LSB units)

12-bit output: 032h (200d in LSB units)

### List of Black Level Adjustment Register

| Register        | Register details |       | Initial value | Setting value |
|-----------------|------------------|-------|---------------|---------------|
|                 | Address          | bit   |               |               |
| BLKLEVEL [11:0] | 30DCh            | [7:0] | 032h          | 000h to 3FFh  |
|                 | 30DDh            | [3:0] |               |               |

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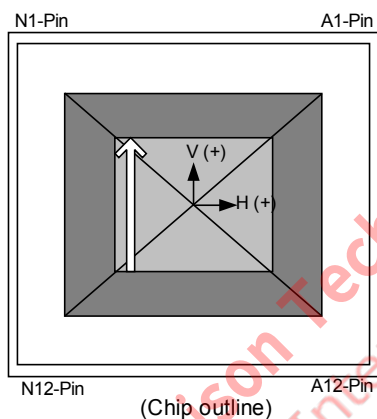
## Normal Operation and Inverted Operation

The sensor readout direction (normal / inverted) in vertical direction can be switched by VREVERSE register and in horizontal direction can be switched by HREVERSE register. See the section of “Image Data Output Format” at datasheet for the order of readout lines in normal and inverted modes, and for other register settings. If the vertical readout direction is switched during streaming, one invalid frame occurs, while regarding the horizontal readout direction switching, no invalid frame occurs.

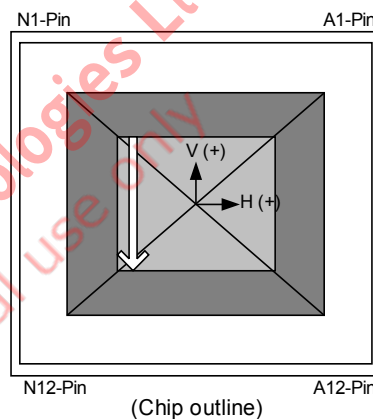
### List of Drive Direction Setting Register

| Register | Register details |     | Initial value | Setting value              |
|----------|------------------|-----|---------------|----------------------------|
|          | Address          | bit |               |                            |
| HREVERSE | 3020h            | [0] | 0h            | 0h: Normal<br>1h: Inverted |
| VREVERSE | 3021h            | [0] | 0h            | 0h: Normal<br>1h: Inverted |

In normal mode

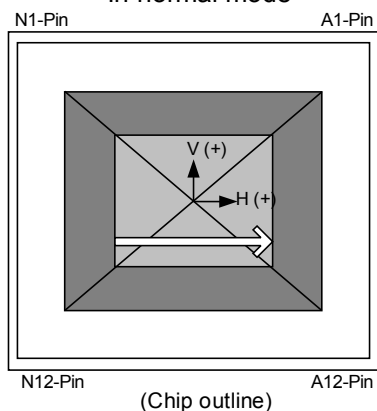


In inverted mode

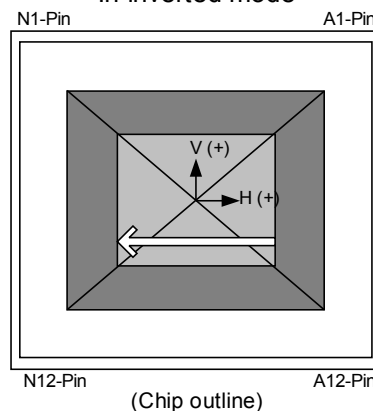


Normal and Inverted Drive Outline in Vertical Direction (TOP VIEW)

In normal mode



In inverted mode



Normal and Inverted Drive Outline in Horizontal Direction (TOP VIEW)

## Shutter and Integration Time Settings

This sensor has a variable electronic shutter function that can control the integration time in line units. In addition, this sensor performs rolling shutter operation in which electronic shutter and readout operation are performed sequentially for each line.

Note) For integration time control, an image which reflects the setting is output from the frame after the setting changes.

### Example of Integration Time Setting

The sensor's integration time is obtained by the following formula.

$$\text{Integration time} = (1\text{H period}) \times (1 \text{ frame period} - \text{SHR0}) + T_{\text{offset}}$$

- \*1 The frame period is determined by the input XVS when the sensor is operating in slave mode, or the register VMAX value in master mode. The frame period is designated in 1H units, so the time is determined by (Number of lines  $\times$  1H period).
- \*2 See "Operating mode" for the 1H period.
- \*3  $T_{\text{offset}} = 0$

In this section, the shutter operation and storage time are shown as in the figure below with the time sequence on the horizontal axis and the vertical address on the vertical axis. For simplification, shutter and readout operation are noted in line units.

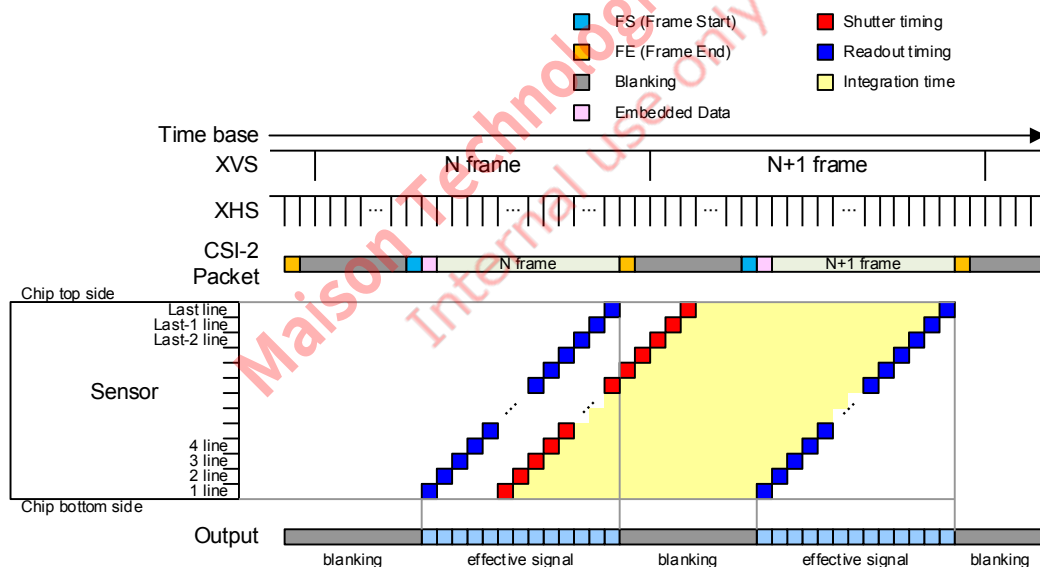


Image Drawing of Shutter Operation

## Normal Exposure Operation (Controlling the Integration Time in 1H Unit)

The integration time can be controlled by varying the electronic shutter timing. In the electronic shutter settings, the integration time is controlled by the SHR0 [19:0] register. Set SHR0 [19:0] to a value between 3 and (Number of lines per frame - 1). When the sensor is operating in slave mode, the number of lines per frame is determined by the XVS interval (number of lines), using the input XHS interval as the line unit.

When the sensor is operating in master mode, the number of lines per frame is determined by the VMAX register. The number of lines per frame differs according to the operating mode.

### Registers Used to Set the Integration Time in 1H Unit

| Register    | Register details |       | Initial value | Setting value                                                                                                               |
|-------------|------------------|-------|---------------|-----------------------------------------------------------------------------------------------------------------------------|
|             | Address          | bit   |               |                                                                                                                             |
| SHR0 [19:0] | 3050h            | [7:0] | 00006h        | Sets the shutter sweep time.<br>3 to (Number of lines per frame - 1)<br>* Others: Setting prohibited                        |
|             | 3051h            | [7:0] |               |                                                                                                                             |
|             | 3052h            | [3:0] |               |                                                                                                                             |
| VMAX [19:0] | 3028h            | [7:0] | 008CAh        | Sets the number of lines per frame<br>(only in master mode). See "Operating<br>mode" for the setting value in each<br>mode. |
|             | 3029h            | [7:0] |               |                                                                                                                             |
|             | 302Ah            | [3:0] |               |                                                                                                                             |

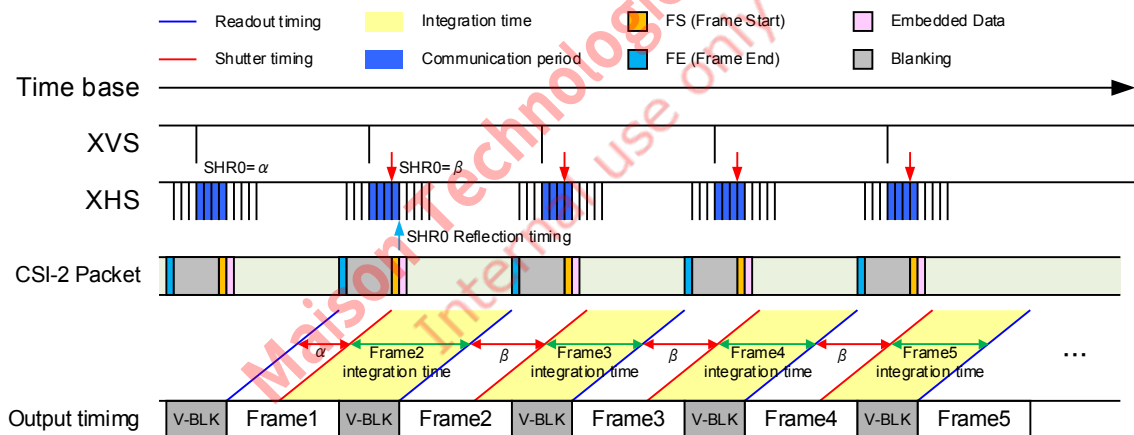


Image Drawing of Integration Time Control within a Frame

### Long Exposure Operation (Control by Expanding the Number of Lines per Frame)

Long exposure operation can be performed by lengthening the frame period.

When the sensor is operating in slave mode, this is done by lengthening the input vertical sync signal (XVS) pulse interval.

When the sensor is operating in master mode, it is done by designating a larger register VMAX [19:0] value compared to normal operation. When the integration time is extended by increasing the number of lines, the rear V blanking increases by an equivalent amount.

When set to a number of V lines or more than that noted for each operating mode, the imaging characteristics are not guaranteed during long exposure operation.

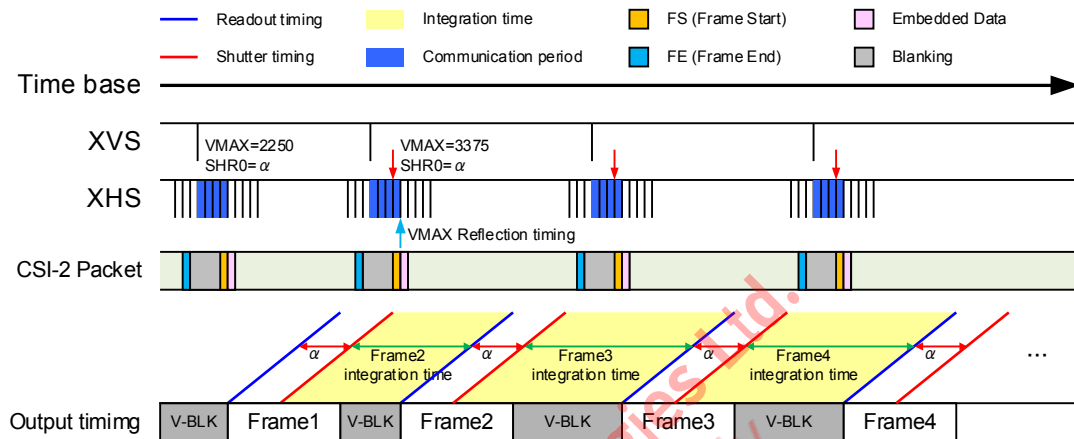


Image Drawing of Long Integration Time Control by Adjusting the Frame Period

### Example of Integration Time Settings

The example of register setting for controlling the storage time is shown below.

#### Example of Integration Time Settings

| Operation           | Sensor setting (register) |        | Integration time                   |
|---------------------|---------------------------|--------|------------------------------------|
|                     | VMAX*                     | SHR0** |                                    |
| All-pixel scan mode | 2250                      | 2249   | 1H + T <sub>offset</sub>           |
|                     |                           | ⋮      | ⋮                                  |
|                     |                           | N      | (2250 - N) H + T <sub>offset</sub> |
|                     |                           | ⋮      | ⋮                                  |
|                     |                           | 3      | 2247H + T <sub>offset</sub>        |

\* In sensor master mode. In slave mode, the interval is the same as XVS input.

\*\* The SHR0 setting value (N) is set between "3" and "the VMAX value – 1".

\*\*\* T<sub>offset</sub> = 0

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## Signal Output

### CSI-2 output

The output formats of this sensor supports the following modes.

CSI-2 serial 2 Lane / 4 Lane / 8Lane / 4Lane × 2ch, RAW10 / RAW12

The 2 Lane / 4 Lane / 8 Lane / 4 Lane × 2ch serial signal output method using this sensor is described below.

Complied with the CSI-2, data is output using 2 Lane / 4 Lane. The image data is output from the CSI-2 output pin.

The DMO1P / DMO1N are called the Lane1 data signal, the DMO2P / DMO2N are called the Lane2 data signal, the

DMO3P / DMO3N are called the Lane3 data signal, the DMO4P / DMO4N are called the Lane4 data signal, the

DMO5P / DMO5N are called the Lane5 data signal, the DMO6P / DMO6N are called the Lane6 data signal, the

DMO7P / DMO7N are called the Lane7 data signal, the DMO8P / DMO8N are called the Lane8 data signal.

In addition, the clock signals are output from DCK1P / DCK1N / DCK2P / DCK2N of the CSI-2 pins.

About 2 Lane / 4 Lane / 8 Lane, Use a clock signal of DCK1P / DCK1N.

About 4 Lane × 2ch, as for Lane 1 to 4, DCK1P / DCK1N, Lane 5 to 8 use a clock signal of DCK2P / DCK2N.

In 2 Lane mode, data is output from Lane1 and Lane2.

In 4 Lane mode, data is output from Lane1, Lane2, Lane3 and Lane4.

In 8 Lane mode, data is output from Lane1, Lane2, Lane3, Lane4, Lane5, Lane6, Lane7 and Lane8.

In 4 Lane × 2ch mode, data is output divide a horizontal direction from side to side, and the left side outputs the Lane1,2,3,4 right side from Lane5,6,7,8.

The bit rate maximum value is 2376 Mbps.

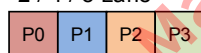
The select of RAW10 / RAW12 is set by the register: MDBIT [1:0]. The number of output lanes is set by the register:

LANEMODE [2:0]. Unused lanes output signals conformed to MIPI standard.

| Register       | Register details |       | Initial value | Setting value                                          |
|----------------|------------------|-------|---------------|--------------------------------------------------------|
|                | Address          | bit   |               |                                                        |
| MDBIT [1:0]    | 3023h            | [1:0] | 1h            | 0h: RAW10<br>1h: RAW12                                 |
| LANEMODE [2:0] | 3040h            | [2:0] | 3h            | 1h: 2lane<br>3h: 4lane<br>6h: 4lane × 2ch<br>7h: 8lane |

The formats of RAW12 and RAW10 are shown below.

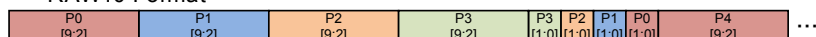
2 / 4 / 8 Lane



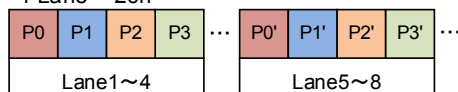
→ RAW12 Format



→ RAW10 Format



4 Lane × 2ch



→ RAW12 Format



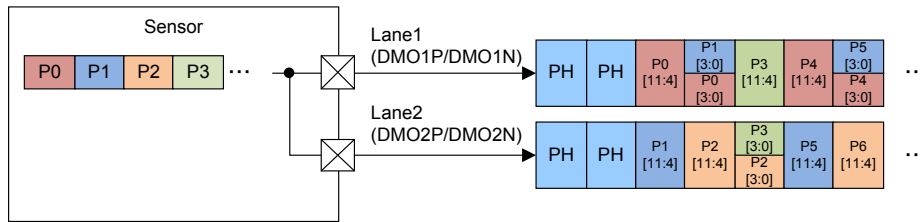
→ RAW10 Format



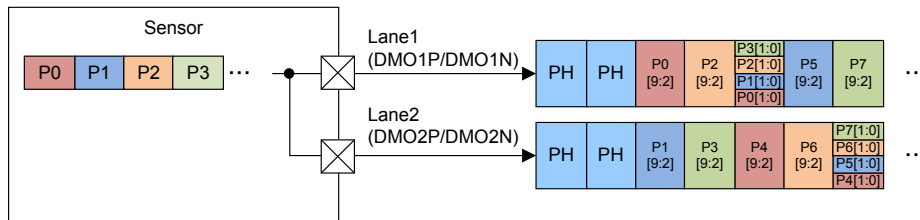
The Example of Format of RAW12 / RAW10

The each format of 2 Lane, 4 Lane, 4Lane × 2ch and 8 Lane are shown below.

a) 2 Lane-RAW12

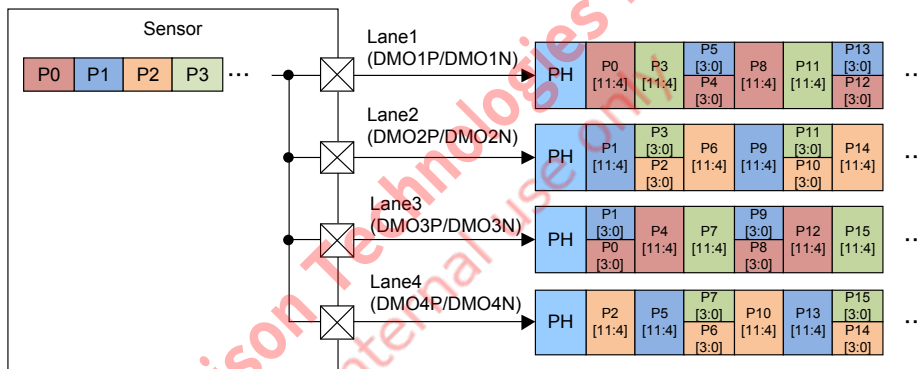


b) 2 Lane-RAW10

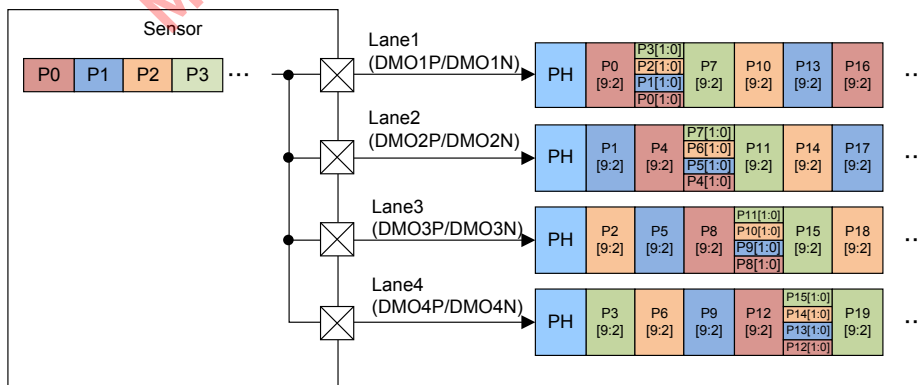


## 2 Lane Output Format

c) 4 Lane-RAW12

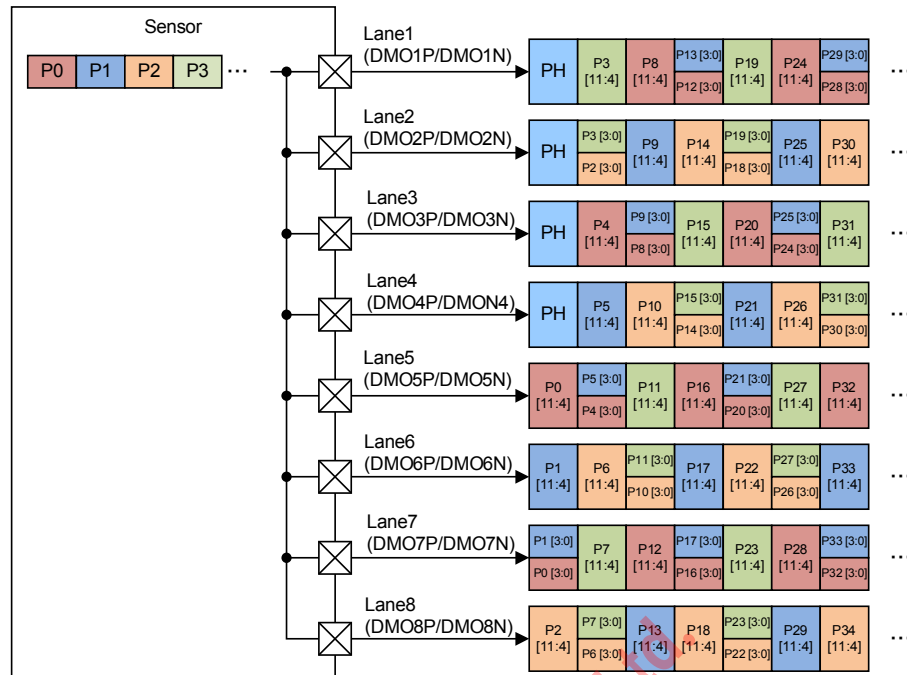


d) 4 Lane-RAW10

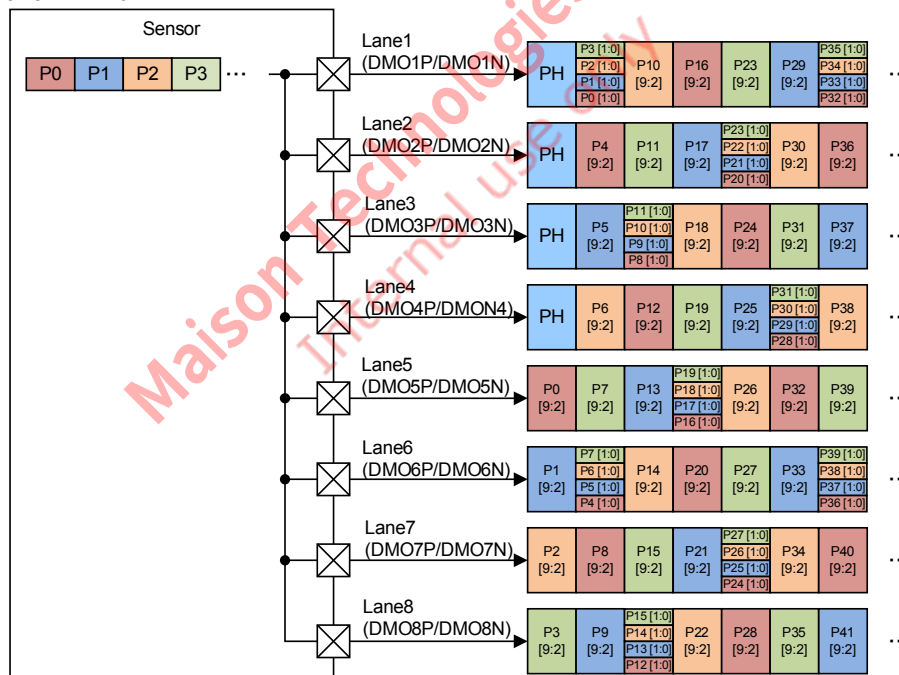


## 4 Lane Output Format

e) 8 Lane-RAW12

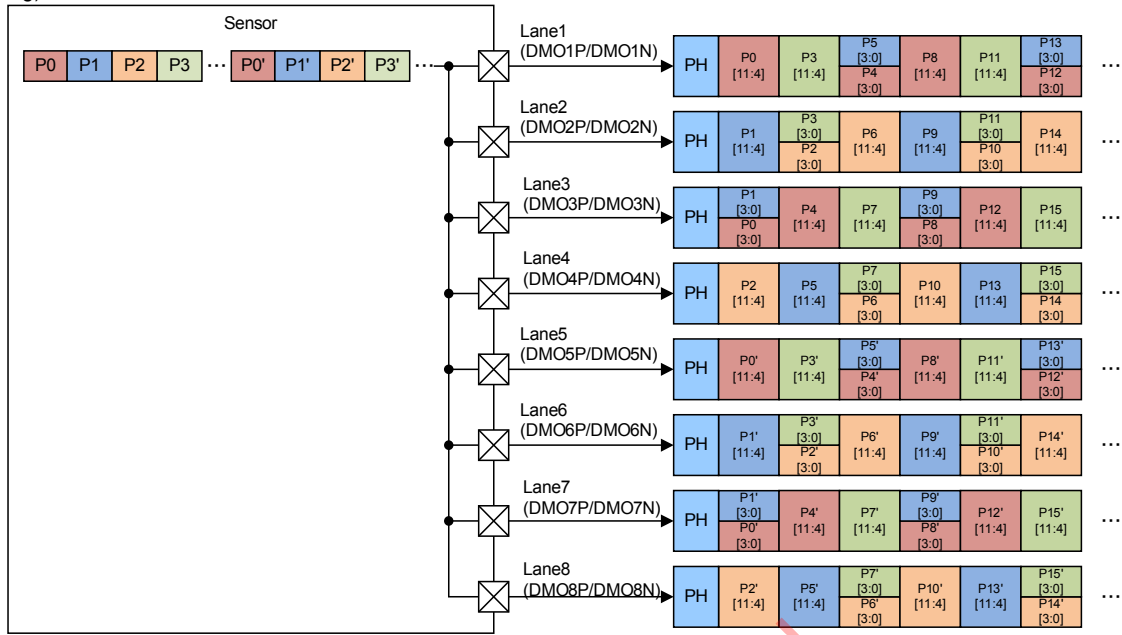


f) 8 Lane-RAW10

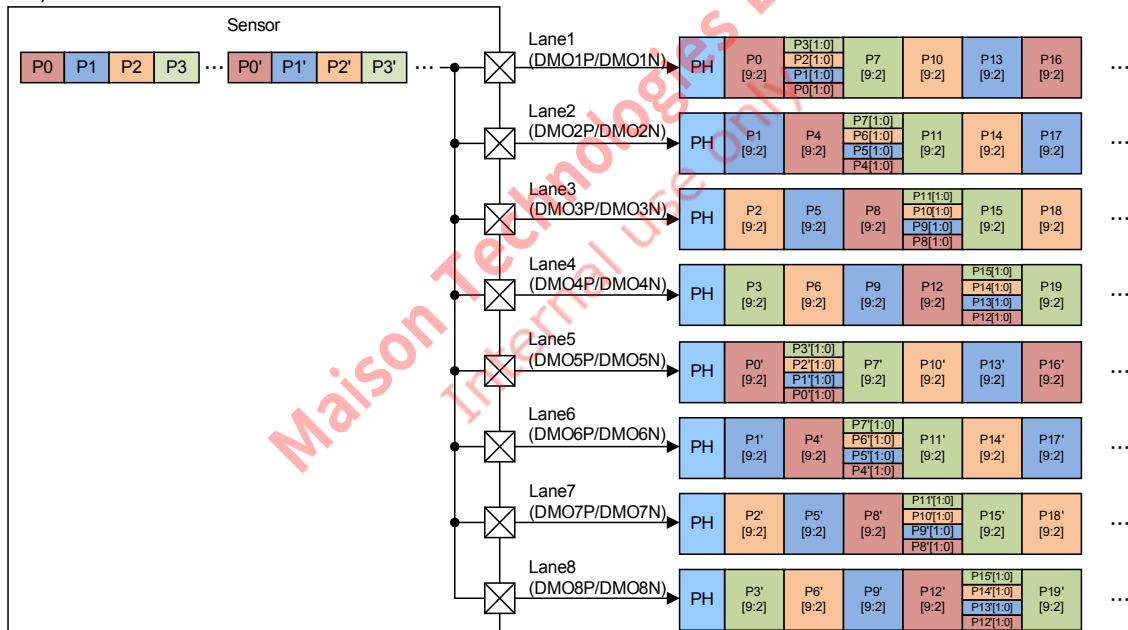


8 Lane Output Format

g) 4 Lane × 2-RAW12



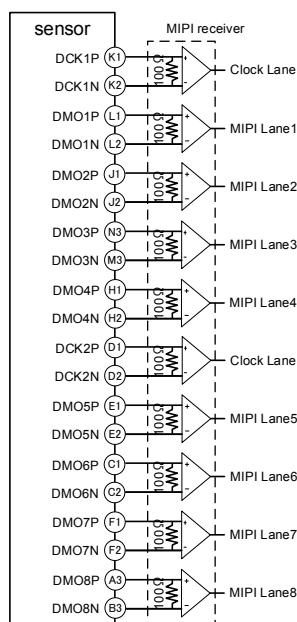
h) 4 Lane × 2-RAW10



4 Lane × 2ch Output Format

## MIPI Transmitter

Output pins (DMO1P, DMO1N, DMO2P, DMO2N, DMO3P, DMO3N, DMO4P, DMO4N, DMO5P, DMO5N, DMO6P, DMO6N, DMO7P, DMO7N, DMO8P, DMO8N, DCK1P, DCK1N, DCK2P, DCK2N) are described in this section.



Relationship between Pin Name and MIPI Output Lane

The pixel signals are output by the CSI-2 High-speed serial interface.

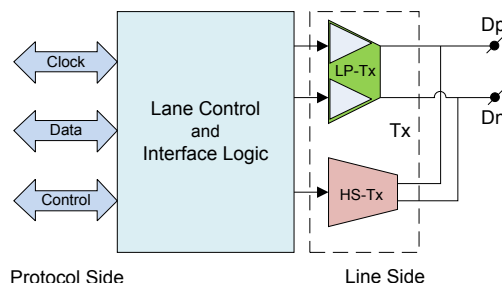
See the MIPI Standard

- MIPI Alliance Standard for Camera Serial Interface 2 (CSI-2) Version 1.2
- MIPI Alliance Specification for D-PHY Version 1.2

The CSI-2 transfers one bit with a pair of differential signals. The transmitter outputs differential current signal after converting pixel signals to it. Insert external resistance in differential pair in a series or use cells with a built-in resistance on the Receiver side. When inserting an external resistor, as close as possible to the Receiver. The differential signals maintain a constant interval and reach the receiver with the shortest wiring length possible to avoid malfunction. The maximum bit rate of each Lane is as follows.

About 2 Lane and 4 Lane, 2376 Mbps / Lane

About 8 Lane and 4 Lane × 2ch, 1440 Mbs / Lane.



Universal Lane Module Functions

### Number of Internal A/D Conversion Bits Setting

The number of internal A/D conversion bits can be selected from 10 bits or 11 bits by the register ADBIT [1:0]. See the chapter of "Operating mode" for the correspondence with each mode. The actual operation is 11 bits, but it is described as 12 bit in the Datasheet and AppNote.

#### List of Bit Width Selection

| Register    | Register details |       | Initial value | Setting value                                          |
|-------------|------------------|-------|---------------|--------------------------------------------------------|
|             | Address          | bit   |               |                                                        |
| ADBIT [1:0] | 3022h            | [1:0] | 1h            | 0h: AD 10bit<br>1h: AD 12bit (11 bit + digital dither) |

### Output Signal Range

In CSI-2 output mode, the sensor output has either a 10 bit, 12 bit gradation, and the maximum output value is the 3FFh value (10 bit output), the FFFh one (12 bit output).

The output range for each output gradation is shown in the table below.

#### Output Gradation and Output Range (CSI-2 Output)

| Output gradation | Output value |      |
|------------------|--------------|------|
|                  | Min.         | Max. |
| 10 bit           | 000h         | 3FFh |
| 12 bit           | 000h         | FFFh |

## INCK Setting

The available operation mode varies according to INCK frequency. When input either 13.5 MHz, 18 MHz, 24 MHz, 27 MHz, 36 MHz, 37.125 MHz, 72 MHz or 74.25 MHz for INCK frequency, Set INCK\_SEL.  
The INCK setting register shown in the table below.

### INCK Setting Register

| Register | Register details |       | Initial value | Setting value                                                                                                         |
|----------|------------------|-------|---------------|-----------------------------------------------------------------------------------------------------------------------|
|          | Address          | Bit   |               |                                                                                                                       |
| INCK_SEL | 3014h            | [3:0] | 0h            | 0h: 74.25 MHz<br>1h: 37.125 MHz<br>2h: 72 MHz<br>3h: 27 MHz<br>4h: 24 MHz<br>5h: 36 MHz<br>6h: 18 MHz<br>7h: 13.5 MHz |

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### Global Timing setting

The table below shows the setting value of Global Timing available with about change "Data Rate".  
When data rate is either 525 Mbps, 594 Mbps, 720 Mbps, 891 Mbps, 1050 Mbps, 1188 Mbps, 1440 Mbps, 1782 Mbps, 2079 Mbps or 2376 Mbps for data rate, Set DATARATE\_SEL. Global timing is set automatically.

#### Data Rate Setting Register

| Register     | Register details |       | Initial value | Setting value                                                                                                                                                                           |
|--------------|------------------|-------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|              | Address          | Bit   |               |                                                                                                                                                                                         |
| DATARATE_SEL | 3015h            | [3:0] | 2h            | Data rate setting<br>0h: 2376 Mbps<br>1h: 2079 Mbps<br>2h: 1782 Mbps<br>3h: 1440 Mbps<br>4h: 1188 Mbps<br>5h: 891 Mbps<br>6h: 720 Mbps<br>7h: 594 Mbps<br>8h: 525 Mbps<br>9h: 1050 Mbps |

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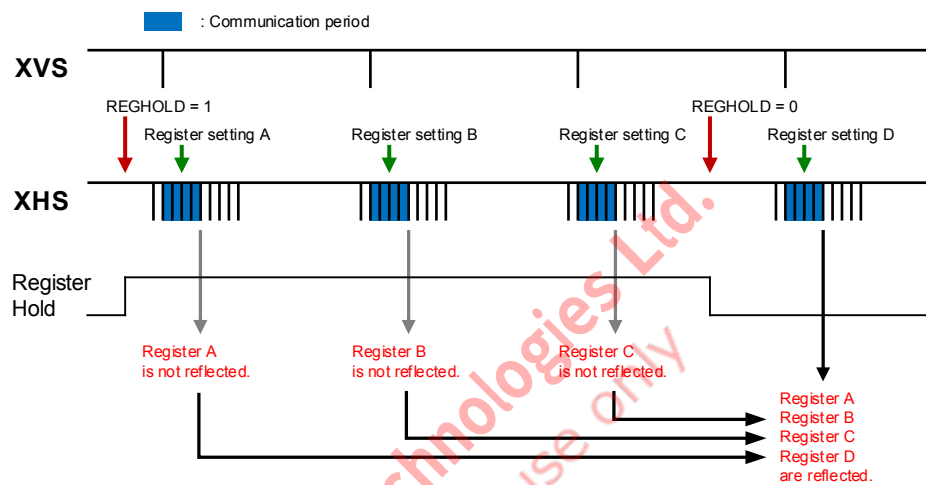


## Register Hold Setting

V reflected register setting can be transmitted with divided to several frames and it can be reflected globally at a certain frame by the register REGHOLD. Setting REGHOLD = 1 prevents the registers that set thereafter from being reflected at the frame reflection timing. The registers that are set when setting REGHOLD = 1 are reflected globally by setting REGHOLD = 0 at the desired frame to reflect the register.

### Register Hold Setting Register

| Register | Register details |     | Initial value | Setting value                          |
|----------|------------------|-----|---------------|----------------------------------------|
|          | Address          | bit |               |                                        |
| REGHOLD  | 3001h            | [0] | 0h            | 0: Invalid<br>1: Valid (Register hold) |



Register Hold Setting

## Mode Transitions

The Mode transition between operations is shown below. These examples shown in case that setting is completed within one communication timing.

List of Mode Transition

| Transition                                                                                                                                                                                                            |    |                               | State                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------------------------------|-----------------------------------------------------------------------|
| Horizontal direction normal                                                                                                                                                                                           | -> | Horizontal direction inverted | Via the Standby state is unnecessary.                                 |
| Horizontal direction inverted                                                                                                                                                                                         | -> | Horizontal direction normal   |                                                                       |
| All-pixel scan mode                                                                                                                                                                                                   | -> | Window cropping mode          | Via the Standby state is unnecessary. One invalid frame is generated. |
| Window cropping mode                                                                                                                                                                                                  | -> | All-pixel scan mode           |                                                                       |
| Vertical direction normal                                                                                                                                                                                             | -> | Vertical direction inverted   |                                                                       |
| Vertical direction inverted                                                                                                                                                                                           | -> | Vertical direction normal     |                                                                       |
| Vertical direction line number change<br>(Master mode: VMAX change, Slave mode: XVS interval change) *1                                                                                                               |    |                               |                                                                       |
| Horizontal direction 1H period change<br>(Master mode: HMAX change, Slave mode: XHS interval change)                                                                                                                  |    |                               |                                                                       |
| - Transition between modes other than above<br>- Change the input frequency of INCK *2<br>- Change the register setting noted “S” in the reflection timing column of the IMX678_Standard_Register_Setting excel file. |    |                               | Via the standby state is necessary.                                   |

\*1 When the changing Vertical derection line number, the output lines are the setting value, but output image is the valid image which is set before. And the image of new setting is output at next frame.

\*2 When changing input INCK frequency, care should be taken not to be input pulses whose width are shorter than the High / Low level width in front and behind of the INCK pulse at the frequency change. If the pulses above generate at the frequency change, change INCK frequency during system reset in the state of XCLR = Low, and then perform system clear in the state of XCLR = High following the item of "Power on sequence" in the section of "Power on / off sequence". Execute initial setting again because the register settings become default state after system clear.

\*3 When changing input output lane number between 8 Lane and 4 Lane x 2ch, via XCLR = Low,

## XSIZE\_OVERLAP Setting

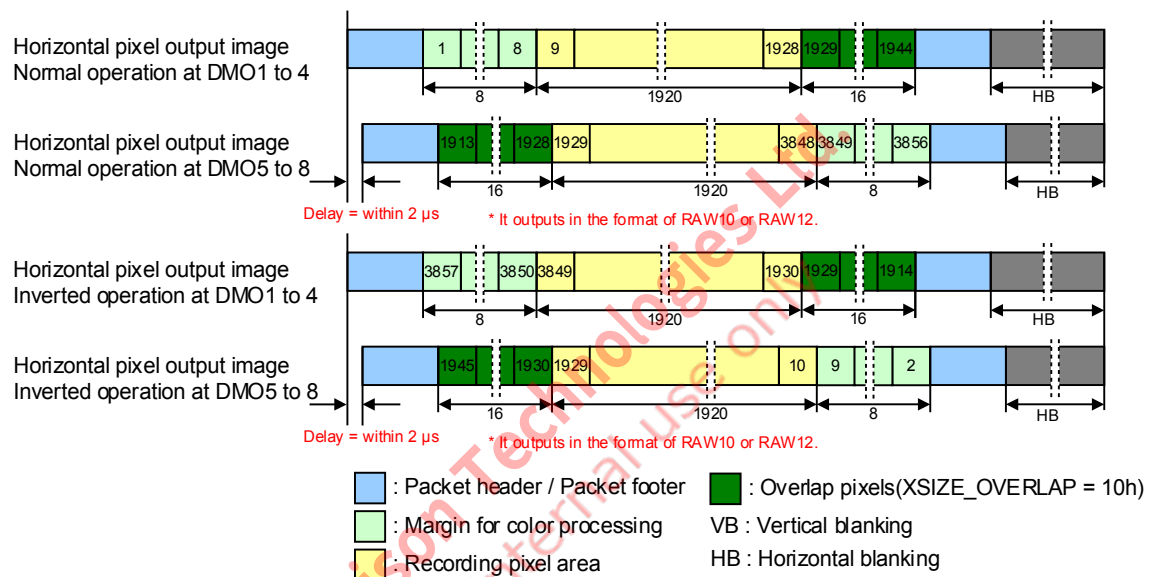
The XSIZE\_OVERLAP register can be duplicated for H pixel several minutes when set XSIZE\_OVERLAP [10:0] and can output the frame of right and left.

Use with values shown below is recommended.

Effective only in "4Lane × 2ch".

### XSIZE\_OVERLAP Register

| Register             | Register details |       | Initial value | Setting value                                                                 |
|----------------------|------------------|-------|---------------|-------------------------------------------------------------------------------|
|                      | Address          | Bit   |               |                                                                               |
| XSIZE_OVERLAP [10:0] | 3042h            | [7:0] | 000h          | 000h to 3F0<br>Multiples of 16 (MAX: 1920d (780h))<br>Smaller than PIX_HWIDTH |
|                      | 3043h            | [2:0] |               |                                                                               |



The Example of Format of XSIZE\_OVERLAP

## Revision History

| Date of change | Revision | Page | Contain of Change                                                                                                                                                                                                                                                                                                                                                             |
|----------------|----------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2021/09/01     | 0.1      | -    | First Edition                                                                                                                                                                                                                                                                                                                                                                 |
| 2021/12/24     | 1.0      | 3    | Fixd Operating mode                                                                                                                                                                                                                                                                                                                                                           |
|                |          | 12   | Correction: the description of Normal Operation and Inverted Operation                                                                                                                                                                                                                                                                                                        |
|                |          | 13   | Correction: Calculation formula of integration time<br>Integration time = 1 frame period - SHR0 × (1H period) + TOffset<br>-> Integration time = (1H period) × (1 frame period – SHR0) + TOffset<br>Delete *3 Set multiple of 2 about the time of SHR0.                                                                                                                       |
|                |          | 14   | Correction: Controlling the Integration Time<br>“2H Units” -> “1H Unit”<br>SHR0: 8 to (Number of lines per frame - 2)<br>-> 3 to (Number of lines per frame - 1)<br>Delete “* Set value multiple of 2”                                                                                                                                                                        |
|                |          | 16   | Correction: Example of Integration Time Settings<br>SHR0: 2248, 2N, 8 -> 2249, N, 3<br>Integration time: 2H, (2250 - 2N) H, 2242H -> H, (2250 - N) H, 2247H<br>** The SHR0 setting value (N) is set between “8” and “the VMAX value (M) – 2”.<br>-> ** The SHR0 setting value (N) is set between “3” and “the VMAX value – 1”.                                                |
|                |          | 21   | Correction: comments in the schematic                                                                                                                                                                                                                                                                                                                                         |
|                |          | 24   | Correction: DATARATE_SEL Initial Value 4h -> 2h                                                                                                                                                                                                                                                                                                                               |
| 2022/03/10     | 2.0      | 27   | Correction: the figure of "The Example of Format of XSIZE_OVERLAP" to match the Drive Timing Chart of 4Lane × 2ch in the Datasheet                                                                                                                                                                                                                                            |
|                |          | 3, 4 | Correction: Operating mode<br>Review of supportable conditions                                                                                                                                                                                                                                                                                                                |
| 2022/04/19     | 3.0      | 3, 4 | Correction: Operating mode<br>Correction of minimum operating data rate for 4lane/10bit/60fps and 4lane/12bit/50fps from 1782 Mbps to 1440 Mbps.<br>Correction of minimum operation data rate for 4lane/12bit/60fps from 2079 Mbps to 1792 Mbps.<br>Added 594 Mbps/Lane as minimum Data rate.<br>Correction of the Data rate 525 Mbps/Lane to work only at INCK 36 or 72 MHz. |
|                |          | 5    | Correction: Typos<br>“Dual Gain” -> “Clear HDR”<br>Delete “Multiple exposure HDR”                                                                                                                                                                                                                                                                                             |
|                |          | 6    | Correction: Typo<br>PIX_VST Register details: [4:0] -> [3:0]                                                                                                                                                                                                                                                                                                                  |
|                |          | 8    | Correction: Typo<br>HMAX [15:0] Initial value: 0226h -> 044Ch                                                                                                                                                                                                                                                                                                                 |
|                |          | 11   | Correction: Typos<br>BLKLEVEL [9:0] -> [11:0], Register details: [1:0] -> [3:0]                                                                                                                                                                                                                                                                                               |
|                |          | 16   | Correction: Typo<br>“H + T <sub>offset</sub> ” -> “1H + T <sub>offset</sub> ”                                                                                                                                                                                                                                                                                                 |

| Date of change | Revision | Page | Contain of Change                                                                                                                                                                                                             |
|----------------|----------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2022/07/19     | 4.0      | 6    | Correction: Restrictions on Window cropping mode<br>Description of "PIX_HST, PIX_HWIDTH"<br>( $n_4 \geq 64$ ) -> ( $n_4 > 64$ , $n_4 \neq 128$ )<br>Added description of "Where $n_1$ to 4 are integer equal or more than 0." |

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